Dwijesh Dookraz

SOFTWARE ENGINEER · BACKEND · AI SYSTEMS · APPLIED MACHINE LEARNING

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SUMMARY

Software Engineer with First Class Honours (BSc Computer Science, University of Southampton), specialising in backend infrastructure and applied machine learning. At Nusmark, designed production Python APIs, calendar synchronisation services, and NLP-driven automation pipelines deployed on Google Cloud Platform (GCP). Research experience in computer vision and deep learning — achieving Pearson r = 0.87 in rPPG-based heart rate estimation from facial video without contact sensors.

TECHNICAL SKILLS

Languages	Python, TypeScript, JavaScript
Frameworks	FastAPI, Flask, React, Next.js 15
AI & ML	PyTorch, scikit-learn, OpenCV, NumPy, Optuna, NLP, Deep Learning
Cloud & Infra	Google Cloud Platform (GCP), Cloud Run, Firestore, Vercel
Tools	Git, GitHub, OAuth2, Playwright, Sentry, Monaco Editor

EXPERIENCE

Nusmark — *2024 – Present*

Backend Engineer

- Engineered a bidirectional Google Calendar and Outlook synchronisation service using Python, FastAPI, and OAuth2 — handling webhook subscriptions, sync token refresh, and transactional Firestore writes to eliminate duplicate calendar events in production
- Developed a WhatsApp AI assistant using Python, Flask, and WhatsApp Cloud API with NLP-driven task extraction across text, audio, and image pipelines — deployed on Google Cloud Platform (GCP) Cloud Run for automated event creation and user onboarding
- Built a multi-channel notification service using Python, Firestore, and Google Calendar API integrated with Microsoft Graph API, routing reminders and mobile push notifications across cross-provider event lifecycles

PROJECTS

rPPG Heart Rate Estimation · Final-Year Dissertation — *2025*

- Implemented a DeepPhys dual-stream attention CNN using PyTorch and OpenCV to estimate heart rate from facial video without contact sensors — achieving Pearson r = 0.87 and MAE 8.85 BPM on the UBFC-rPPG benchmark
- Collected a 15-subject dataset spanning Fitzpatrick skin types II–VI, combined with UBFC-rPPG in Python and NumPy training pipelines to address demographic bias — recovering cross-population correlation from r = −0.23 to r = +0.59
- Applied Optuna hyperparameter search with subject-aware k-fold cross-validation in PyTorch, benchmarking against ICA, PCA, frequency-domain, and band-pass signal-processing baselines

Hybrid Recommender System — *2026*

- Built matrix factorisation and hybrid collaborative filtering (user-CF + item-CF + content signals) from scratch using Python and NumPy with streaming SGD, Nesterov momentum, and Optuna Bayesian search — achieving MAE 0.587 on 20M MovieLens ratings without loading the full dataset into RAM

VS Code Portfolio · dwijesh.dev — *2024 – 2026*

- Built a browser-based VS Code shell using Next.js 15, TypeScript, and React with Monaco Editor — integrating GitHub API via server-side route handlers, Sentry error monitoring, Vercel Analytics, and a Playwright UI test suite for production deployment
- Designed a proxy API route in Next.js to server-side fetch shields.io and skillicons.dev badges, bypassing browser Content Security Policy restrictions for dynamic badge rendering in the editor preview

Gene Expression Analysis — Osteosarcoma — *2025*

- Built an ML pipeline on GSE1000 microarray data using Python and scikit-learn — ANOVA feature selection across 22,283 probes, Shrinkage LDA with LOOCV (balanced accuracy 0.80), Random Forest (2,000 trees), and k-means clustering (k=3) to confirm 26 ECM genes downregulated at 32h via GO enrichment

EDUCATION

University of Southampton *2021 – 2024*

BSc Computer Science · First Class Honours

Dissertation: **Machine Learning-Based Heart Rate Measurement Using rPPG — 82%**

Machine Learning Technologies **80%** Social Computing Techniques **83%** Cloud Application Development **79%**
Software Engineering Group Project **73%**